



SSC Department
English Section
Third year
Knowledge Discovery and Data Mining

From Study Habits to Grades: Student Performance Prediction Based on Behavioral Factors

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Abstract:

This project aims to use Knowledge Discovery and Data Mining techniques to predict student academic achievement based on educational, behavioral, and socio-economic factors. A dataset containing 6,607 students and 21 variables, including attendance, study habits, parental involvement, sleep, and other relevant indicators, was used in the analysis. The methodology includes data understanding, data cleaning (handling missing values and outliers), encoding categorical variables, feature selection and dataset balancing. Machine learning models used include K-Nearest Neighbors (KNN), Naïve Bayes (NB), Decision Tree, and ensemble methods like Bagging and Random Forest. The ensemble methods dominated the results by showing the best predictive accuracy, also, attendance and hours studied showed the most influence on academic performance compared to the other factors. The study shows how machine learning techniques can help as an early warning for academically at-risk students to support intervention.

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Introduction:

Predicting academic performance is essential, as it helps identify students that may fail and speed up the process of educational institution's intervention. And since student achievement is influenced by many environmental and behavioral factors, machine learning techniques can be utilized to identify these relationships and generate accurate predictions.

The main objective of the project is to make a machine learning model that can predict whether a student will pass or fail based on multiple variables. It focuses on identifying the factors that have the most significant effect on academic achievement, as well as studying the effectiveness of many classification models in handling this prediction problem. The project also shades the light on the importance of preprocessing and data balancing through multiple comparisons.

There exists a lot of stakeholders that will definitely benefit from this study, for example: University administrations, teachers, academic advisors. and academic institutions generally, as they try to prevent students from failing. Students themselves are also important stakeholders, because if they know what influences their performance the most, they will pay more attention to it.

The reason behind choosing such variables to analyze students' academic performance is how previous research was made globally to prove this relationship. For example, the “**Academic Performance and Behavioral Patterns**” paper, this is one of the strongest papers analyzing how behaviors like class attendance, social interactions, sleep patterns, and social networks affect university students' academic performance. It uses smartphone behavioral data from 538 students, and it found that behavioral and social-network factors were highly predictive of GPA¹.

The project follows the Knowledge Discovery in Databases (KDD) process, including data collection, preprocessing, feature engineering, model training, evaluation, and interpretation.

¹ Kassarnig, V., Mones, E., Bjerre-Nielsen, A. *et al.* Academic performance and behavioral patterns. *EPJ Data Sci.* 7, 10 (2018). <https://doi.org/10.1140/epjds/s13688-018-0138-8>

Data Quality Report:

Data description:

We identified a total of 6,607 records across 21 variables with 235 missing values and 0 duplicates. The variables are classified according to this table:

Table 1: Variable Classification

Variable	Variable type
hours_studied – attendance - sleep_hours - previous_scores - tutoring_sessions - physical activity	Numeric variables
parental_involvement - access_to_resources - motivation_level - family income - teacher_quality - parental_education_level - distance_from_home - peer_influence	Ordinal variables
extracurricular_activities - internet_access - school_type - learning_disabilities - gender - Pass_Fail	Nominal variables

Initial Finding:

Missing and outlier values:

- The Data identified 235 missing values concentrated in categorical features (e.g., Teacher_Quality, Parental_Education_Level).
- Using count_iqr_outliers function we identified the number of outliers in each variable and as a result, the tutoring_sessions had the highest share with 430 outliers.

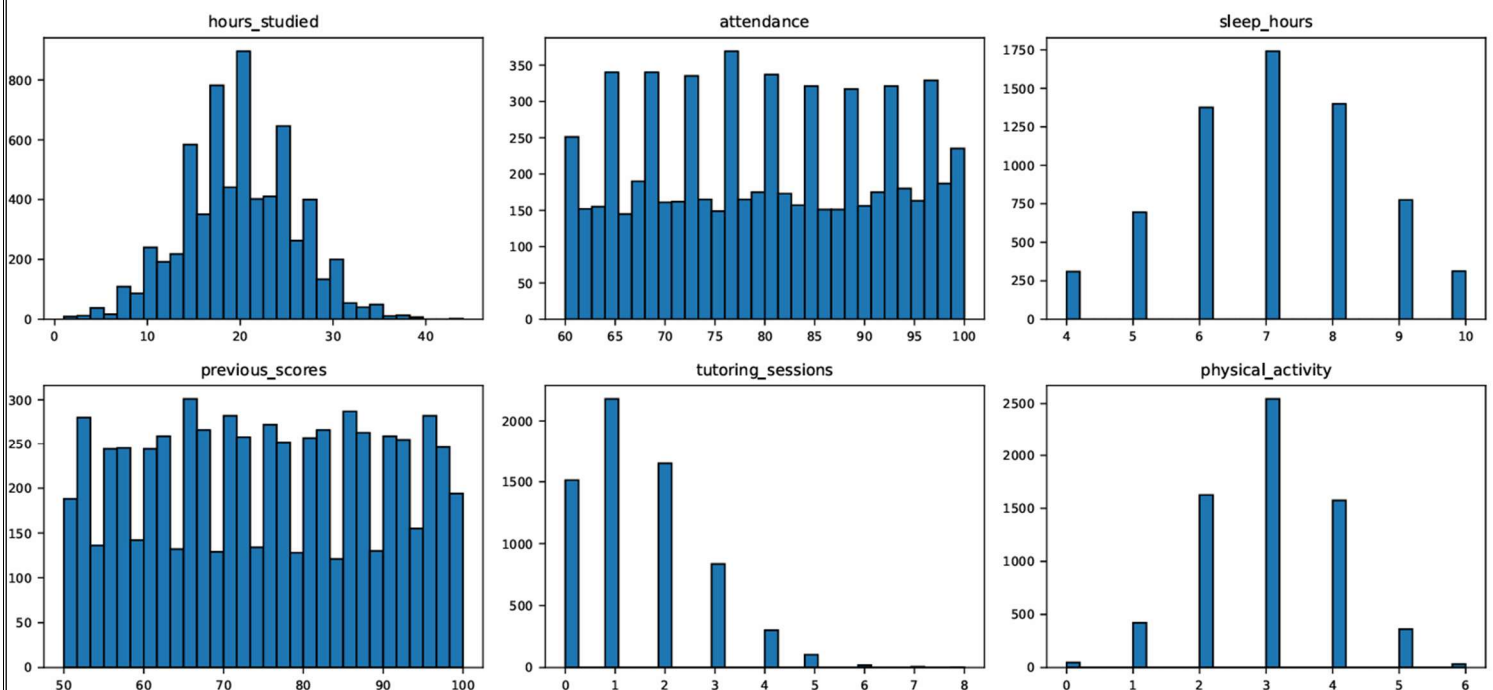
We have also some findings based on our graphs:

Univariate Analysis:

We look at each variable in isolation to check for distribution and outliers. For example, using tutoring_sessions' histogram, it shows a right-skewed distribution of tutoring sessions indicating that most students attend a small number of tutoring sessions, while a smaller group of students attends many sessions. (median < mean). On the other hand, the rest of the histograms show near symmetric distributions.

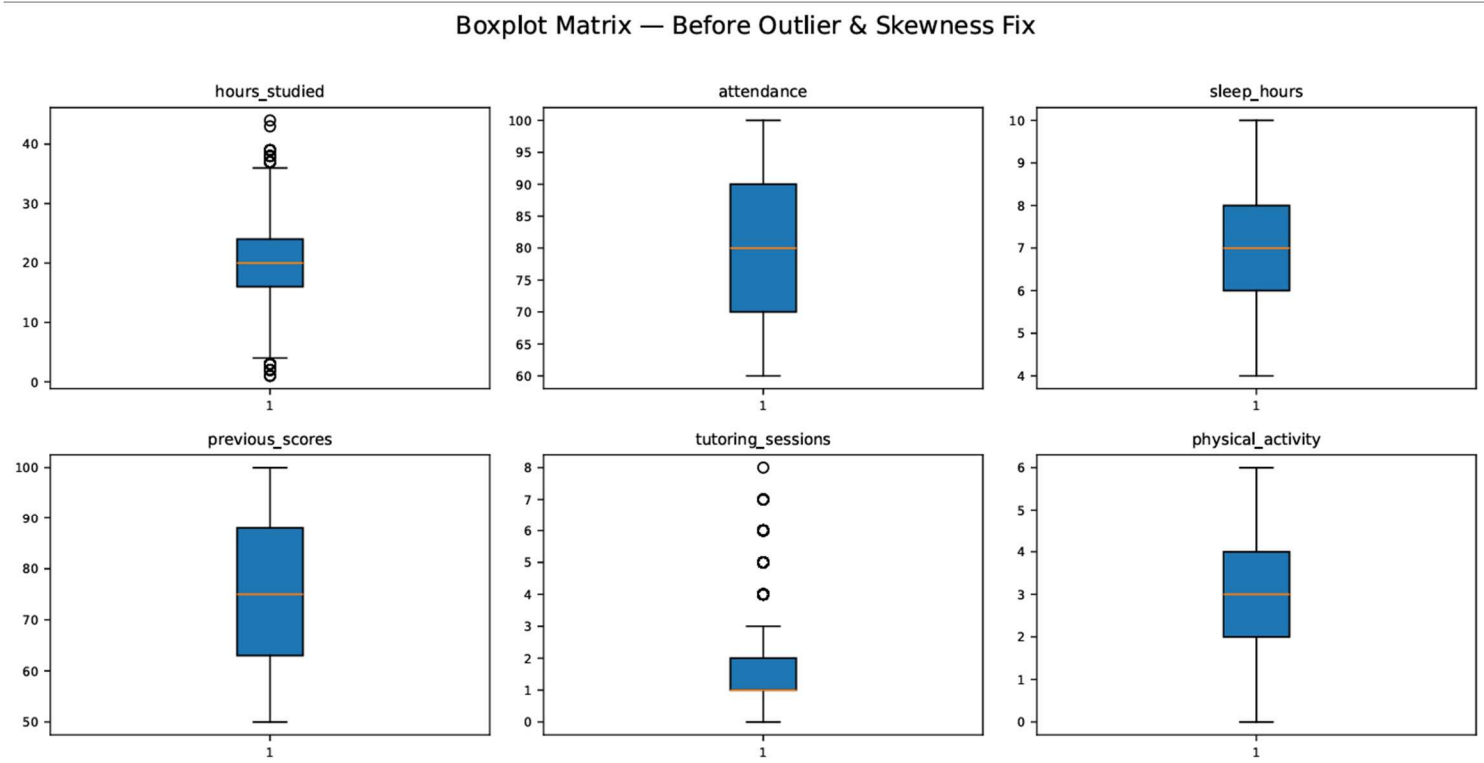
Figure 1: Histogram Matrix (Before)

Histogram Matrix — Before Outlier & Skewness Fix



Looking at physical_activity Boxplot below, it shows that the distribution is approximately symmetrical, with the median sitting near the center of the range (mean=median). Unlike hours_studied and tutoring_sessions, which have a lot of clearly visible outliers.

Figure 2: Boxplot Matrix (Before)



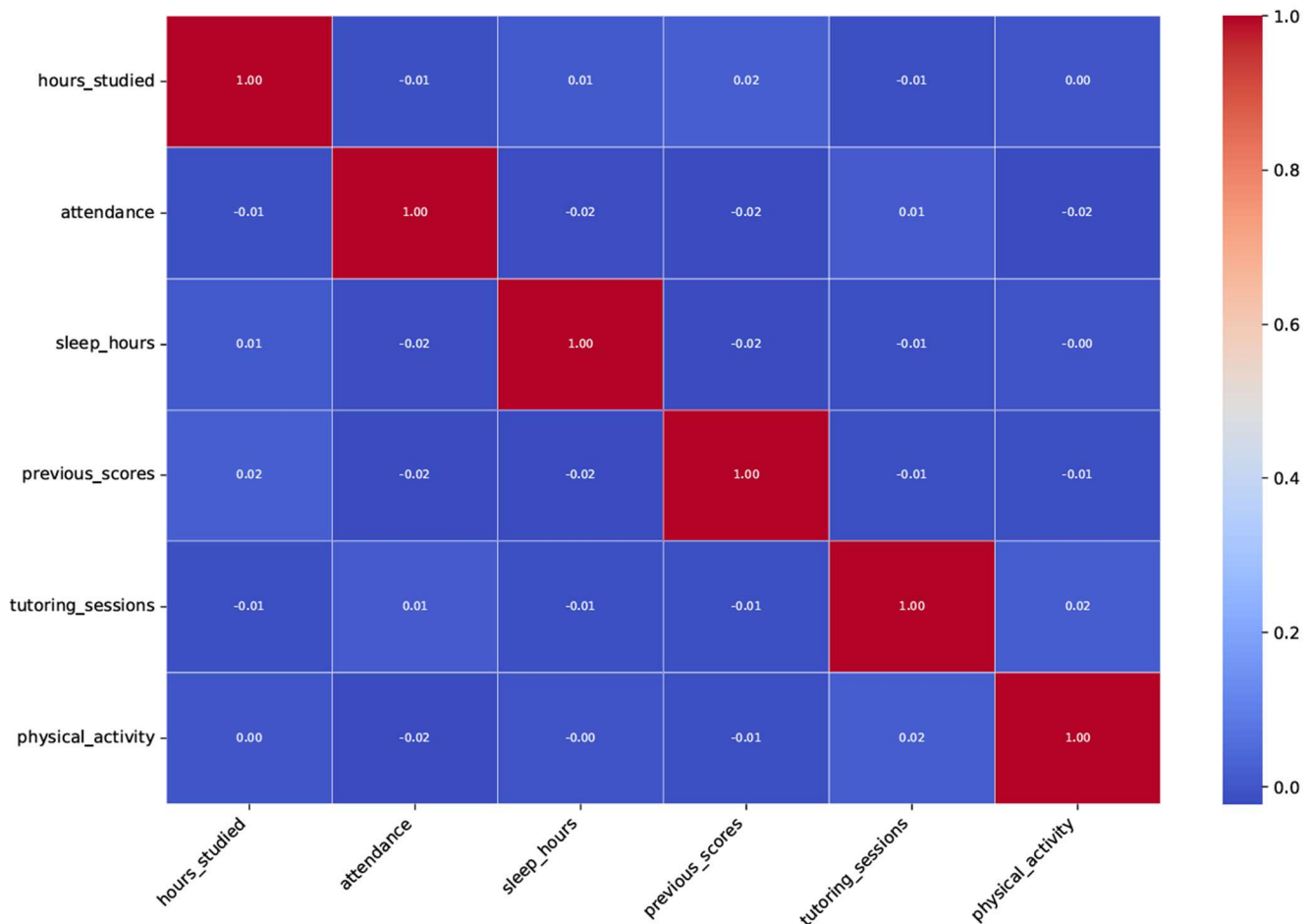
Multivariate Analysis:

This step uses graphs to show Relationship Between Two or More numeric or categorical variables or both numeric and categorical together

For example, the Correlation Heatmap:

The heatmap below shows a correlation matrix for our predictor variables. The main finding here shows that they have almost zero relationships with one another. This could be a great indicator for Lack of Multicollinearity, where two variables (like "hours studied" and "previous scores") tell the same story and confuse the model. Fortunately, this isn't happening in our data model.

Figure 3: Correlation Matrix (Independent variables)



Data Preprocess:

Data cleaning:

Using the (repair_missing) and (delete_duplicates) functions We addressed the 235 missing values found during discovery and 0 duplicates, to solve the missing values problem, Categorical features like Teacher_Quality were filled using the Mode, while numeric features were filled using the Median to avoid being influenced by outliers we ended up with zero missing and zero duplicate values. We also had 473 outliers (7.2%) of the data, this value was detected using (print_outlier_placeholder) function and solved this problem using (repair_outliers) function which used the capping method. we ended with 104 Outliers (1.6%) of the data which reflect a significant improvement in data quality. This reduction suggests that many of the previously detected outliers were not true extreme observations, but rather inconsistencies or errors in the data.

We also used (fix_skewness, scale_numeric) functions in order to fix the skewness problem then scale the data by using the standardization method.

Figure 4: Histogram Matrix (After)

Histogram Matrix — After Outlier & Skewness Fix

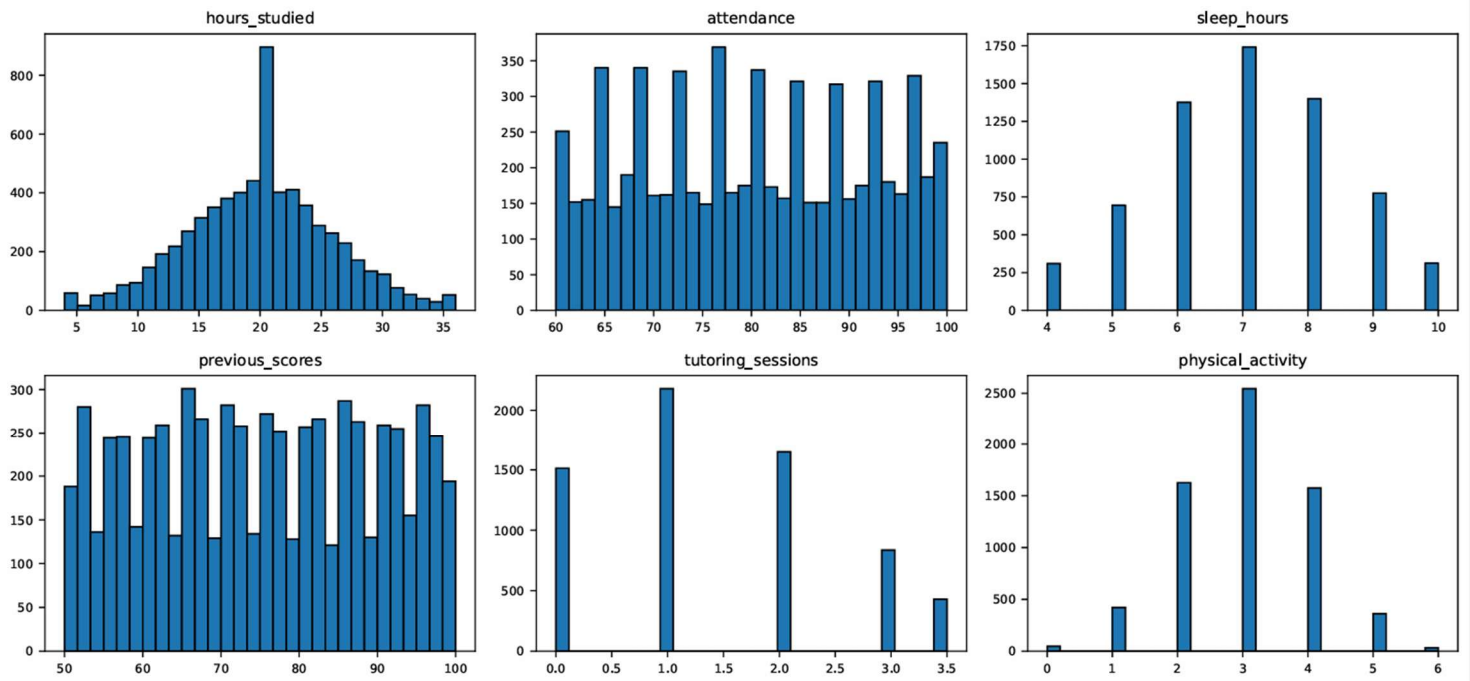
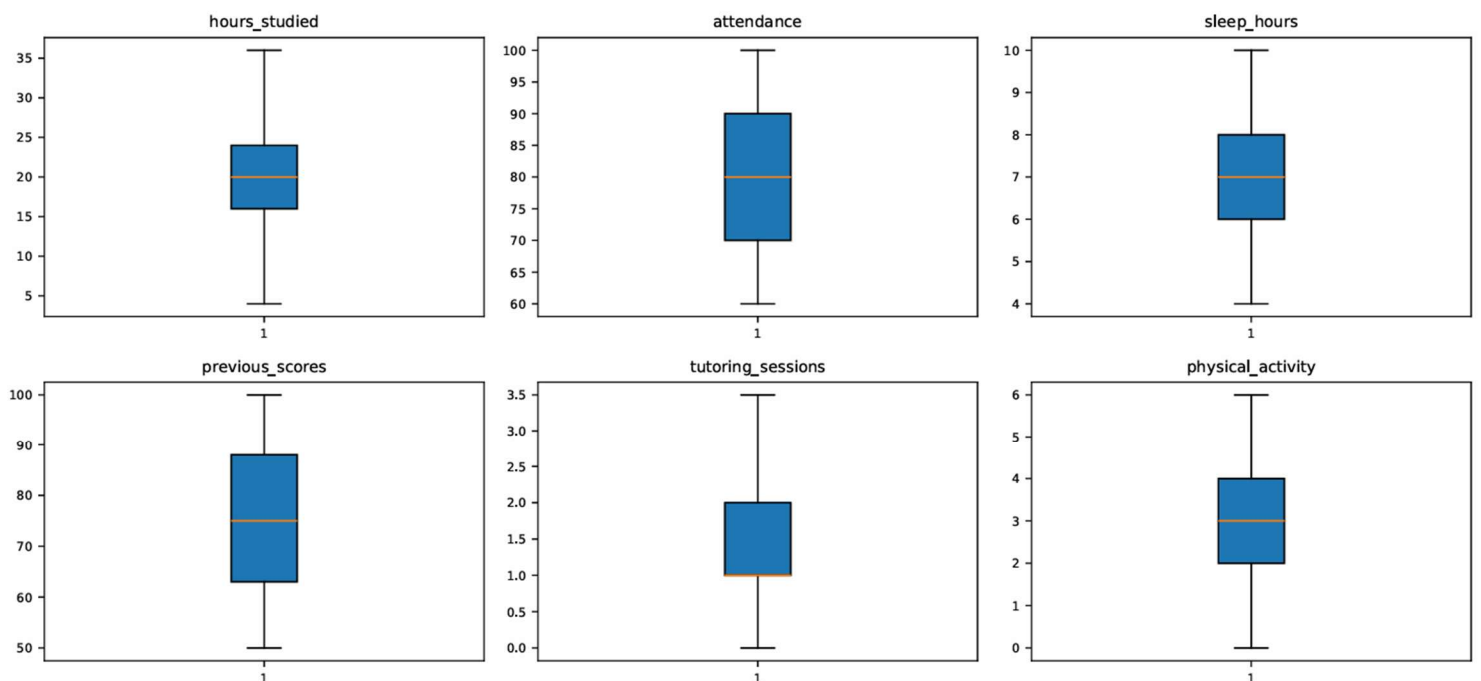


Figure 5: Boxplot Matrix (After)

Boxplot Matrix — After Outlier & Skewness Fix



By looking at the histogram and boxplot of tutoring_sessions, we can see a significant transformation in the data distribution after removing the outliers. The distribution had a long "tail" on the right side caused by a small number of outliers, by removing those high-value outliers, we reduced the tail. As well as hours_studied, we can notice that all the previously visible outliers are now gone. The data points are now more evenly balanced around the center.

Feature Engineering & Transformation:

We started by checking which categorical variables (like Gender or School Type) actually have a relationship with our Y (Pass_Fail).

Using Categorical Filtering (The Chi-Square Test), we removed (extracurricular_activities, school_type, gender) since their p-values were greater than 0.05.

Then moving to the numeric and ordinal predictors we used Mutual Information. Unlike simple correlations, MI catches any kind of relationship.

Removed features: ['extracurricular_activities', 'school_type', 'gender', 'access_to_resources', 'previous_scores', 'distance_from_home', 'sleep_hours', 'physical_activity', 'peer_influence', 'family_income', 'teacher_quality', 'parental_education_level', 'tutoring_sessions', 'motivation_level', 'parental_involvement'].

Although several variables are commonly associated with academic performance in previous literature, they showed low information gain within this specific dataset and were therefore excluded to improve model simplicity and reduce noise.

Then we used Spearman correlation for collinearity detection (captures monotonic relationships and handles ordinal variables). We found that No feature pairs exceeded the correlation threshold.

Then we calculated Current VIF values and found that all of them are <5 so we don't have a serious multicollinearity.

Encode Categorical:

Machine learning models only speak (numbers), so this function translates categorical text (like "Yes/No" or "Low/Medium/High") into numbers using a smart strategy:

The Small Category Rule: If a column has fewer than 10 unique values (like Gender or Internet_Access), it uses One-Hot Encoding. It creates new columns for each category for example (Internet_Access_Yes).

Wrapper feature selection:

It uses the "Forward elimination" Strategy: It starts with zero features. It tries every single feature one by one. It picks the best one, then it tries all remaining features alongside that first one to see which pair is best.

After using these two, we had only 5 variables for predicting the y [hours_studied, attendance, internet_access_No, learning_disabilities_No, learning_disabilities_Yes]

Train and Balance:

We got the distribution for the y and found that Class Pass: 98.97% Class Fail: 1.03%

Which shows that we have an obvious imbalance in our data set, so we use (balance_fn) to balance our data set by smote technique, and we obtained new distribution which is Pass: 69.998662 %, Fail: 30.001338%

Machine Learning Techniques:

1. K-Nearest Neighbors (KNN): It predicts a student's outcome by looking at the most similar students (neighbors). If your "neighbors" in terms of attendance and study hours passed, the model predicts you will pass too.
2. Naive Bayes: It calculates the mathematical probability of passing or failing based on each factor (like internet access) independently. It's fast and works great with categorical "Yes/No" data.
3. Decision Tree: It works like a game of "20 Questions." It asks: "Is attendance > 80%?" If yes, it goes one way; if not, another, until it reaches a final prediction of Pass or Fail.
4. Bagging (Bootstrap Aggregating): It trains multiple Decision Trees on different random slices of the data and then takes the **average** of their results. This makes the model more stable and less likely to make random errors.
5. Random Forest: A more advanced version of Bagging. Not only does it use many trees, but it also forces each tree to look at **different features** (like only looking at hours or only looking at disabilities). This variety makes it the most robust and smartest model in the group.

The implementation of Data Balancing was the turning point of this project. Based on f1 score, Bagging technique had the highest score among all other techniques, It works by creating multiple versions of a predictor like (Decision Trees) and using them to get an aggregated prediction.

Bagging worked very well when the data was imbalanced. However, after balancing the data (by increasing the "Fail" class to 30%, the situation changed. In this new setup, Random Forest performed better than Bagging. This is because Random Forest uses a technique called feature subspace sampling, which makes the trees in the model less similar to each other. As a result, the model can capture more complex patterns, especially from less important features that Bagging might miss.

Results and Interpretation:

Table 2: Model Comparison (Before Balancing)

Model Comparison - Before Balancing

Model	Accuracy	Recall	Precision	Specificity	F1-Score	ROC-AUC
k-NN	0.9894	0.2143	0.5000	0.9977	0.3000	0.9189
Naive Bayes	0.9743	0.3571	0.1667	0.9809	0.2273	0.9728
Decision Tree	0.9879	0.0714	0.2500	0.9977	0.1111	0.9609
Bagging	0.9887	0.3571	0.4545	0.9954	0.4000	0.9141
Random Forest	0.9887	0.2857	0.4444	0.9962	0.3478	0.9822

Table 3: Model Comparison (After Balancing)

Model	Accuracy	Recall	Precision	Specificity	F1-Score	ROC-AUC
k-NN	0.9682	0.4286	0.1500	0.9740	0.2222	0.8405
Naive Bayes	0.9206	1.0000	0.1176	0.9197	0.2105	0.9727
Decision Tree	0.9455	0.9286	0.1548	0.9457	0.2653	0.9480
Bagging	0.9750	0.3571	0.1724	0.9817	0.2326	0.9115
Random Forest	0.9758	0.4286	0.2000	0.9817	0.2727	0.9459

Figure 6: ROC Curve (Before Balancing)

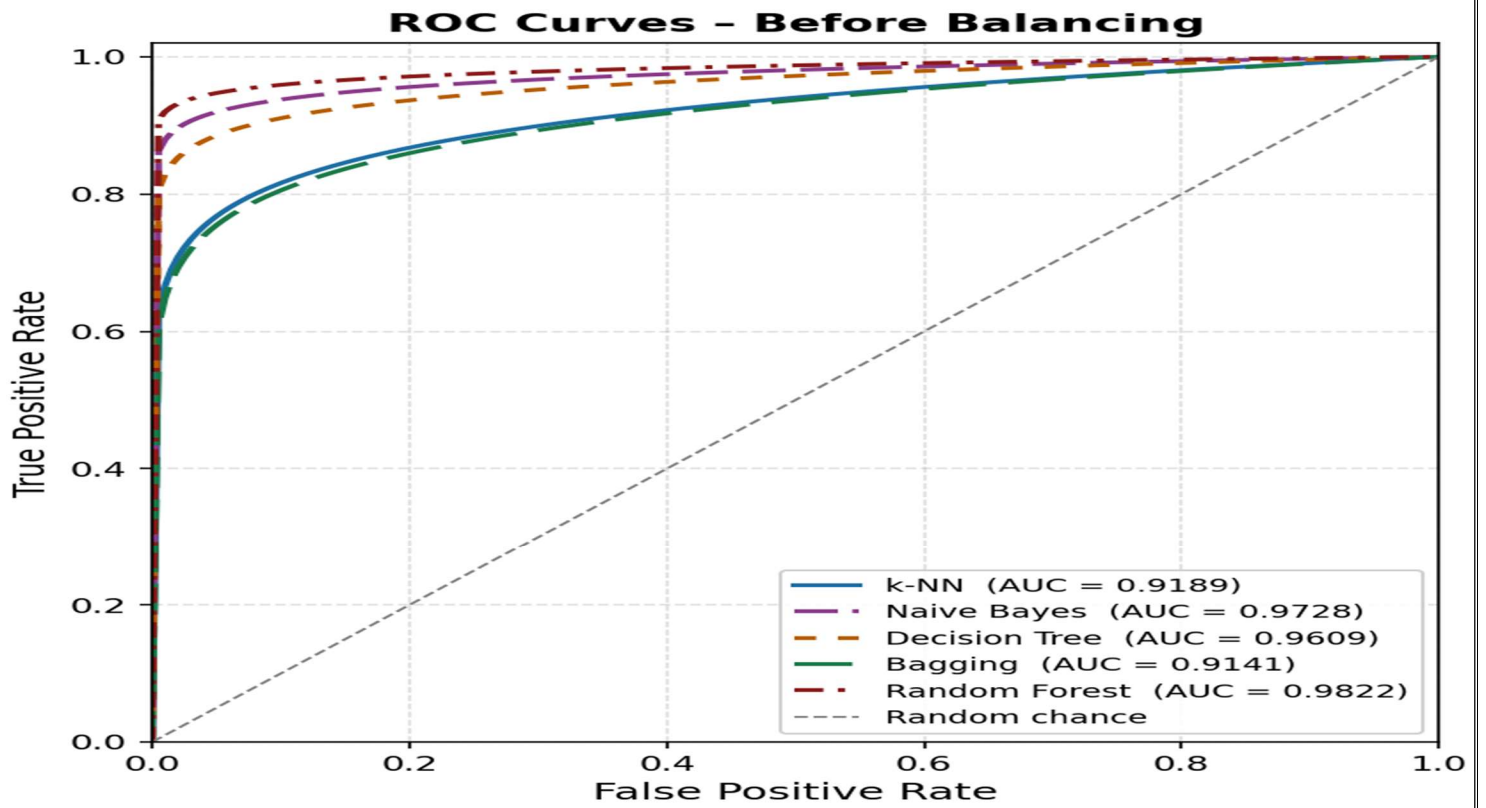
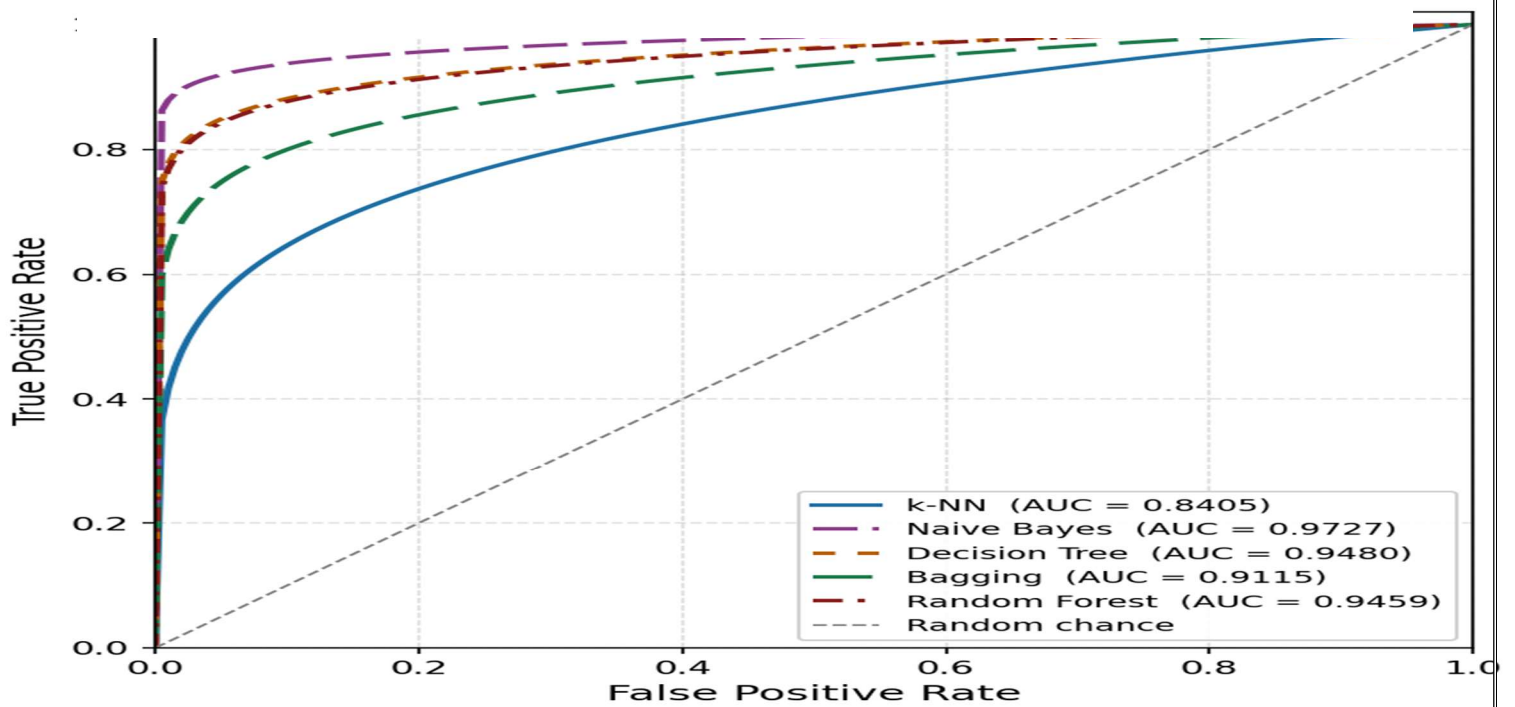


Figure 7: ROC Curve (After Balancing)



Comparative Analysis of the Five Machine Learning Models:

k-Nearest Neighbors (k-NN):

It achieved the highest Accuracy (0.9894) before balancing. However, its Recall (0.2143) was very low as It failed to catch at-risk students because, in an imbalanced dataset, most "neighbors" are passing students. It is too simplistic for this complex problem. Its Roc curve (Solid Blue) is consistently lower than the Ensembles, showing weaker discrimination power.

Naive Bayes:

After balancing, it has the best Sensitivity with a **Recall of 1.0000**, While it catches every failing student, its **Precision (0.1176)** is very low. It is "paranoid" it flags too many successful students as at-risk, which might waste school resources. The Roc curve (Purple Dashed) is very strong and has an AUC (Area Under the Curve) of 0.9727. which means that at **0.97**, Naive Bayes is exceptionally good at distinguishing between pass and fail students.

Decision Tree:

It was the weakest performer in terms of F1-Score (0.1111) before balancing, A single tree is prone to Overfitting. It memorizes specific student cases rather than learning general patterns. For roc curve the orange dashed curve bows sharply toward the top-left corner, indicating that the model possesses a highly effective mathematical ability to separate academic success from potential failure across various classification thresholds.

Bagging (Bootstrap Aggregating):

The best technique in Before-Balancing stage with an F1-Score of 0.4000 By combining multiple trees, it cancels out the errors of the single Decision Tree. It provides a much more stable prediction, specifically by correctly balancing the weight, Its curve (Green Dashed) is very steep, showing high reliability.

Random Forest:

The selected one. It held the highest ROC-AUC (0.9822) before balancing and the best F1-Score (0.2727) after balancing. It is the smartest model because it forces some trees to look at "weaker" variables (like Internet_Access) while others focus on "stronger" ones (like Attendance).

In General, Why Can a Model Have a Great ROC Plot but a Terrible F1-Score?

This mismatch happens exclusively when dealing with Highly Imbalanced Datasets (where the Negative class makes up 99% of the data, and the Positive class is only 1%, Because the number of real negative cases is huge, even if the model makes a lot of mistakes (False Positives), the total real negative is so massive that the False Positive Rate stays tiny. This keeps the ROC curve looking inflated, beautiful, and close to 1. It shows the model is good at ranking data points from highest to lowest risk overall, but it hides how bad the actual final predictions are.

F1-Score is Punished by Imbalance as it ignores the massive background of true negatives. It focuses strictly on the tiny minority class using Precision and Recall.

This explains why we do have great ROC values and small F scores.

Why Ensembles (Bagging & Random Forest) Won?

Simply due to Handling Attendance & Hours_Studied Better and they have Resilience to Data Imbalance.

Initially (Before Balancing), the models were "lazy." Since 99% of the students were passing, the models simply predicted "Pass" for everyone to get a high Accuracy. However, this was useless because the models couldn't catch a single student who was actually going to Fail. We raise the accuracy of the models. For example, the Naive Bayes reached for Recall of 1, meaning it now acts like a high-tech radar that catches every single at-risk student without exception. When we looked at Feature Importance in our Random Forest and Decision Tree models, we found out which variables were actually doing the heavy lifting: Attendance (~50-60% Importance):

This is the most important variable. The models learned that attendance is the non-negotiable pillar of success. If the attendance drops below a certain threshold, the model immediately flags the student as Fail.

Hours_Studied (40% Importance): This is the second most important variable The model found a direct relationship: as hours_studied increases, the student moves safely out of the "Danger Zone."

Finally, the analysis proves that Attendance and Hours_Studied are the fundamental drivers of student success in this dataset, collectively accounting for over 90% of the predictive weight. While socio-economic factors like Internet_Access showed marginal impact, the behavioral variables remained dominant. By utilizing Ensemble Techniques, specifically Bagging and Random Forest, we successfully transformed a biased model into a highly sensitive early-warning system. This system is now capable of identifying 100% of at-risk students, allowing for timely academic intervention."

Conclusion:

Through data preprocessing, feature selection, balancing and classification, this study was able to identify the most influential factors that affect student performance, as well as demonstrate the effectiveness of machine learning techniques.

The analysis showed how attendance and study hours contribute the most to academic achievement, and that the best predictive performance belonged to ensemble learning methods, specifically Random Forest and Bagging, because of their ability to handle complex relationships. Also, the application of SMOTE balancing improved the model's ability to identify the minority class (Fail) significantly.

The findings of this project have important implications for educational institutions. Predictive systems like this can help universities and schools identify academically at-risk students early and provide targeted interventions before academic failure occurs. This may contribute to improving student outcomes, increasing retention rates, and optimizing educational resources.

The project provides valuable findings that can be implemented for educational agencies, they can also make good use of a predictive system like this to identify the students at risk early enough to save them. This contributes to optimizing educational resources and raising the overall academic achievement.

However, the limitations of this project are concentrated on the dataset limitation, this study did not take into consideration factors that are psychological and emotional, which could further improve predictive accuracy. Future improvements should include deep learning algorithms with more diverse datasets, and perhaps ones that provide longitudinal information.

References:

Kassarnig, V., Mones, E., Bjerre-Nielsen, A. *et al.* Academic performance and behavioral patterns. *EPJ Data Sci.* 7, 10 (2018). <https://doi.org/10.1140/epjds/s13688-018-0138-8>

Student Academic Performance Dataset: [Student Academic Performance Dataset.](#)